

STA2005S Regression Practical One

Extension Solutions

Set-up

Libraries

```
library(ggplot2)
library(tibble)
```

Data

```
if (!requireNamespace("DataTidy23RodoSTA2005S")) {
  if (!requireNamespace("remotes", quietly = TRUE)) {
    utils::install.packages("remotes")
  }
  remotes::install_github("MiguelRodo/DataTidy23RodoSTA2005S")
}
```

Load it using the code below:

```
library(tibble) # ensures dataframes print more informatively and concisely than standard
#| echo: true
data("data_tidy_tobacco", package = "DataTidy23RodoSTA2005S")
```

It appears as below:

```
data_tidy_tobacco
```

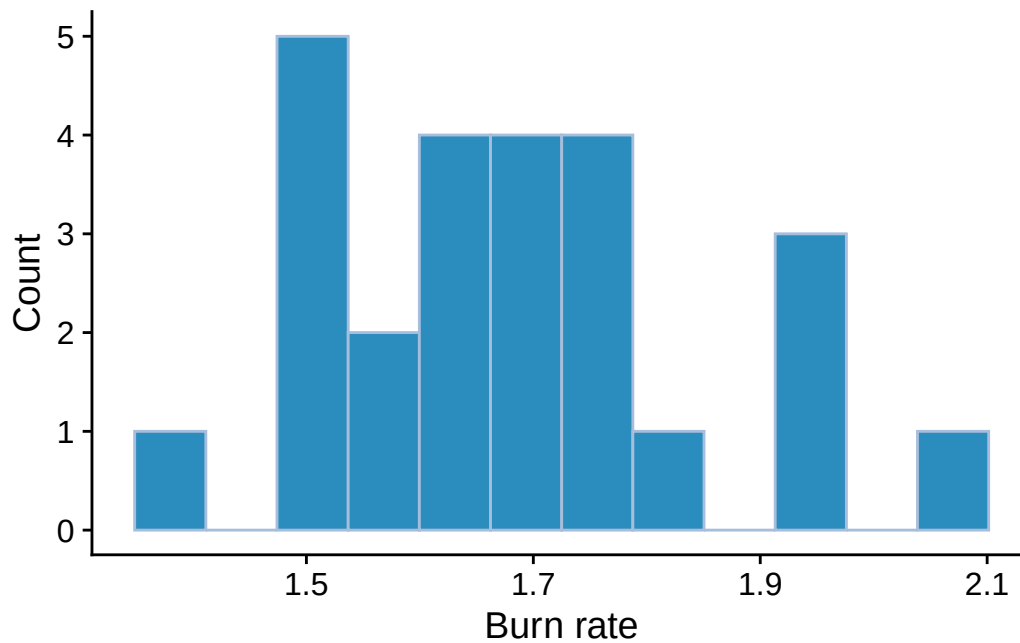
```
# A tibble: 25 x 9
  burn sugar nicotine nitrogen chlorine potas phospho calcium magnes
  <dbl> <dbl>    <dbl>    <dbl>    <dbl> <dbl>    <dbl>    <dbl>    <dbl>
1  1.55  20.0      1.38      2.02      2.9   2.17     0.51     3.47     0.91
2  1.63  12.6      2.64      2.62      2.78  1.72     0.5      4.57     1.25
3  1.66  18.6      1.56      2.08      2.68  2.4      0.43     3.52     0.82
4  1.52  18.6      2.22      2.2       3.17  2.06     0.52     3.69     0.97
5  1.7   14.0      2.85      2.38      2.52  2.18     0.42     4.01     1.12
6  1.68  15.6      1.24      2.03      2.56  2.57     0.44     2.79     0.82
7  1.78  14.5      2.86      2.87      2.67  2.64     0.5      3.92     1.06
8  1.57  18.5      2.18      1.88      2.58  2.22     0.49     3.58     1.01
9  1.6   17.8      1.65      1.93      2.26  2.15     0.56     3.57     0.92
10 1.52  13.4      3.28      2.57      1.74  1.64     0.51     4.38     1.22
# i 15 more rows
```

Questions

Question One

Draw a histogram of the response variable, burn.

```
ggplot(
  data_tidy_tobacco,
  aes(x = burn)
) +
  geom_histogram(col = "#a6bddb", fill = "#2b8cbe", bins = 12) +
  cowplot::theme_cowplot() +
  labs(x = "Burn rate", y = "Count")
```



Question Two

Draw a scatterplot of burn against each of the explanatory variables.

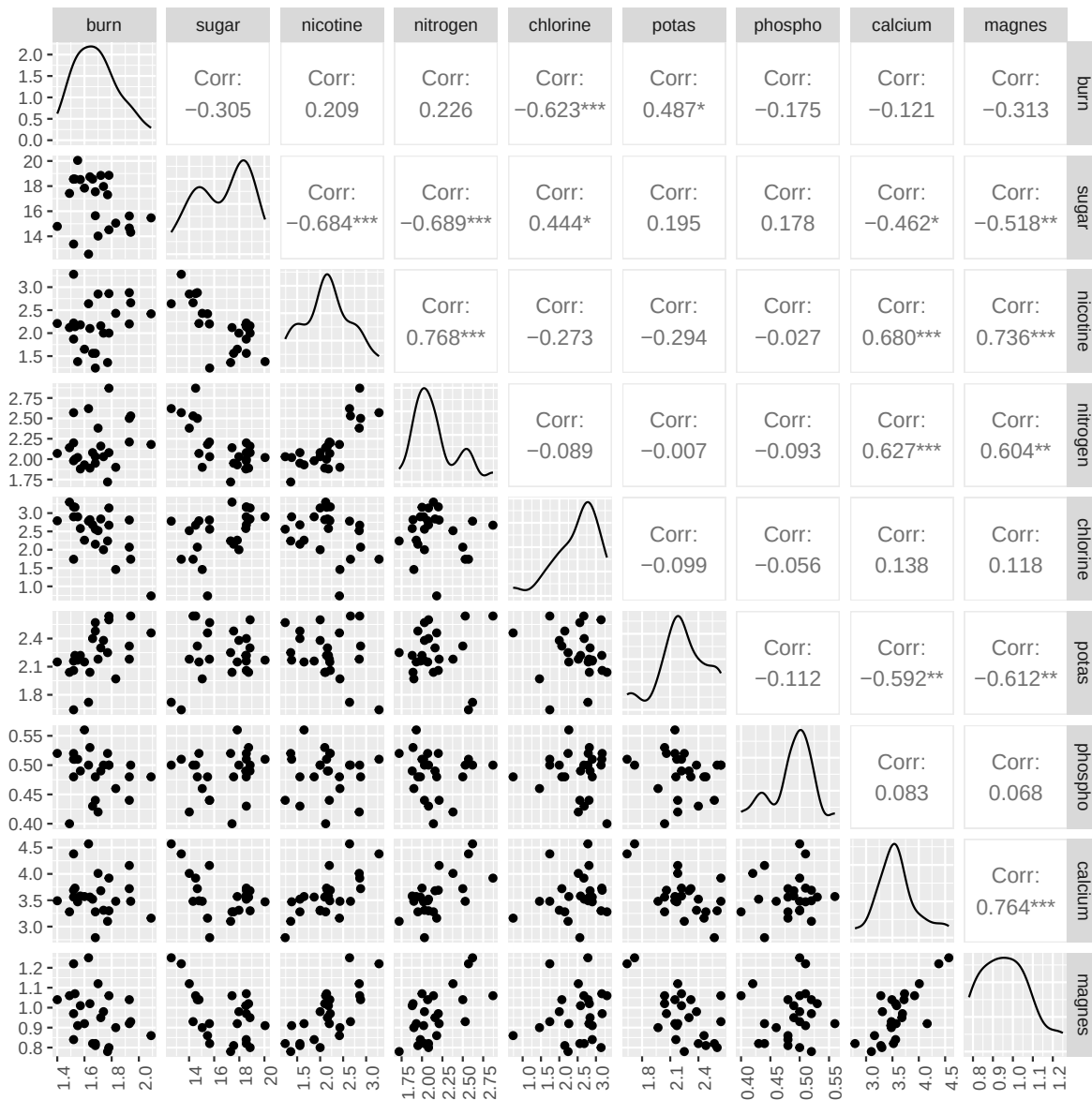
```
p <- GGally::ggpairs(data = data_tidy_tobacco) +
  theme(axis.text.x = element_text(angle = 90))
```

Registered S3 method overwritten by 'GGally':

method from
+.gg ggplot2

```
if (!dir.exists("_tmp")) {
  dir.create("_tmp")
}
path_png_pw <- "_tmp/p-pw.pdf"
ggplot2::ggsave(
  path_png_pw,
  plot = p,
  units = "cm", height = 20, width = 20
)
```

```
pander::pandoc.image(path_png_pw)
```



The plot text is somewhat cramped.

Question Three

Calculate correlation between every possible pair of variables

```
corr_mat <- cor(data_tidy_tobacco)
corr_mat |> signif(2)
```

	burn	sugar	nicotine	nitrogen	chlorine	potas	phospho	calcium	magnes
burn	1.00	-0.30	0.210	0.2300	-0.620	0.4900	-0.180	-0.120	-0.310
sugar	-0.30	1.00	-0.680	-0.6900	0.440	0.1900	0.180	-0.460	-0.520
nicotine	0.21	-0.68	1.000	0.7700	-0.270	-0.2900	-0.027	0.680	0.740
nitrogen	0.23	-0.69	0.770	1.0000	-0.089	-0.0075	-0.093	0.630	0.600
chlorine	-0.62	0.44	-0.270	-0.0890	1.000	-0.0990	-0.056	0.140	0.120
potas	0.49	0.19	-0.290	-0.0075	-0.099	1.0000	-0.110	-0.590	-0.610
phospho	-0.18	0.18	-0.027	-0.0930	-0.056	-0.1100	1.000	0.083	0.068
calcium	-0.12	-0.46	0.680	0.6300	0.140	-0.5900	0.083	1.000	0.760
magnes	-0.31	-0.52	0.740	0.6000	0.120	-0.6100	0.068	0.760	1.000

Question Four

Fit a simple linear regression of burn against potassium.

Part a

Do so from first principles (i.e. not using the `lm` function)

Sub-part i

Estimate β .

```
X_mat <- data_tidy_tobacco[, "potas"] |>
  dplyr::mutate(intercept = 1) |>
  dplyr::select(intercept, potas) |>
  as.matrix()
y_vec <- data_tidy_tobacco[, "burn"] |>
  as.matrix()
beta_vec <- solve(t(X_mat) %*% X_mat) %*% t(X_mat) %*% y_vec
beta_vec |> signif(2)
```

	burn
intercept	0.98
potas	0.32

Sub-part ii

Estimate σ^2 .

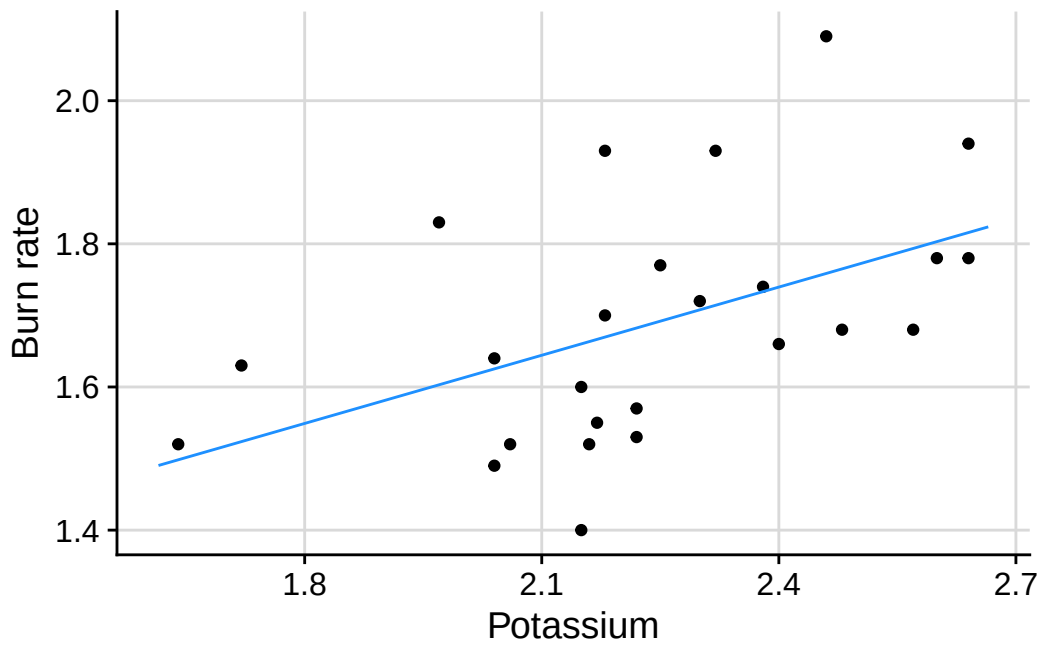
```
n <- nrow(X_mat)
k <- ncol(X_mat)
pred_vec <- X_mat %*% beta_vec
s2 <- sum((y_vec - pred_vec)^2) / (n - k)
s2 |> signif(2)
```

```
[1] 0.022
```

Sub-part iii

Create a plot of the raw data and the fitted model.

```
x_vec <- range(data_tidy_tobacco$potas, max(data_tidy_tobacco$potas))
range_len <- diff(x_vec)
x_vec <- x_vec + c(-0.025, 0.025) * range_len
y_vec <- beta_vec[1] + beta_vec[2] * x_vec
ggplot(
  tibble::tibble(x = x_vec, y = y_vec)
) +
  cowplot::theme_cowplot() +
  cowplot::background_grid(major = "xy") +
  geom_point(mapping = aes(x = potas, y = burn), data = data_tidy_tobacco) +
  geom_line(aes(x = x, y = y), col = "dodgerblue") +
  labs(x = "Potassium", y = "Burn rate")
```



Part b

Do so using built-in R functions (i.e. `lm`).

Sub-part i

Display the summary output (use the `summary` object on the fitted model object above).

```
fit <- lm(burn ~ potas, data = data_tidy_tobacco)
summary(fit)
```

Call:

```
lm(formula = burn ~ potas, data = data_tidy_tobacco)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.26019	-0.11241	-0.02307	0.07806	0.33138

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.9775	0.2677	3.652	0.00133 **

```

potas          0.3175      0.1189    2.671  0.01365 *
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.149 on 23 degrees of freedom
Multiple R-squared:  0.2367,    Adjusted R-squared:  0.2035
F-statistic: 7.134 on 1 and 23 DF,  p-value: 0.01365

```

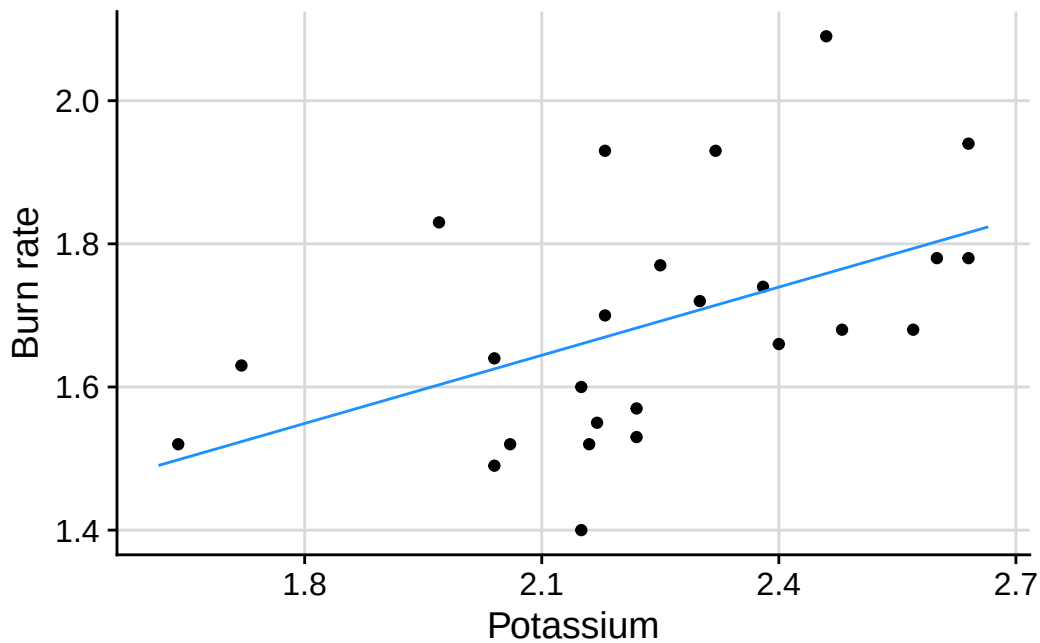
Sub-part ii

Create a plot of the raw data with the model overlaid.

```

beta_vec <- coef(fit) |> as.matrix()
y_vec <- beta_vec[1] + beta_vec[2] * x_vec
ggplot(
  tibble::tibble(x = x_vec, y = y_vec)
) +
  cowplot::theme_cowplot() +
  cowplot::background_grid(major = "xy") +
  geom_point(
    mapping = aes(x = potas, y = burn),
    data = data_tidy_tobacco
  ) +
  geom_line(aes(x = x, y = y), col = "dodgerblue") +
  labs(x = "Potassium", y = "Burn rate")

```

Question Five

Fit a multiple linear regression of `burn` against all the variables

Part a

Do so from first principles (i.e. not using the `lm` function)

Sub-part i

```
X_mat <- data_tidy_tobacco |>
  dplyr::select(sugar:magnes) |>
  dplyr::mutate(intercept = 1) |>
  dplyr::select(intercept, everything()) |>
  as.matrix()
y_vec <- data_tidy_tobacco[, "burn"] |>
  as.matrix()
beta_vec <- solve(t(X_mat) %*% X_mat) %*% t(X_mat) %*% y_vec
beta_vec |> signif(2)
```

	burn
intercept	1.6000
sugar	-0.0042
nicotine	0.0910
nitrogen	-0.0330
chlorine	-0.1300
potas	0.3100
phospho	-0.6200
calcium	0.1800
magnes	-0.6500

Sub-part ii

Estimate σ^2 .

```
n <- nrow(X_mat)
k <- ncol(X_mat)
pred_vec <- X_mat %*% beta_vec
s2 <- sum((y_vec - pred_vec)^2) / (n - k)
s2 |> signif(2)
```

```
[1] 0.012
```

Sub-part iii

Create a plot of the raw data and the fitted model.

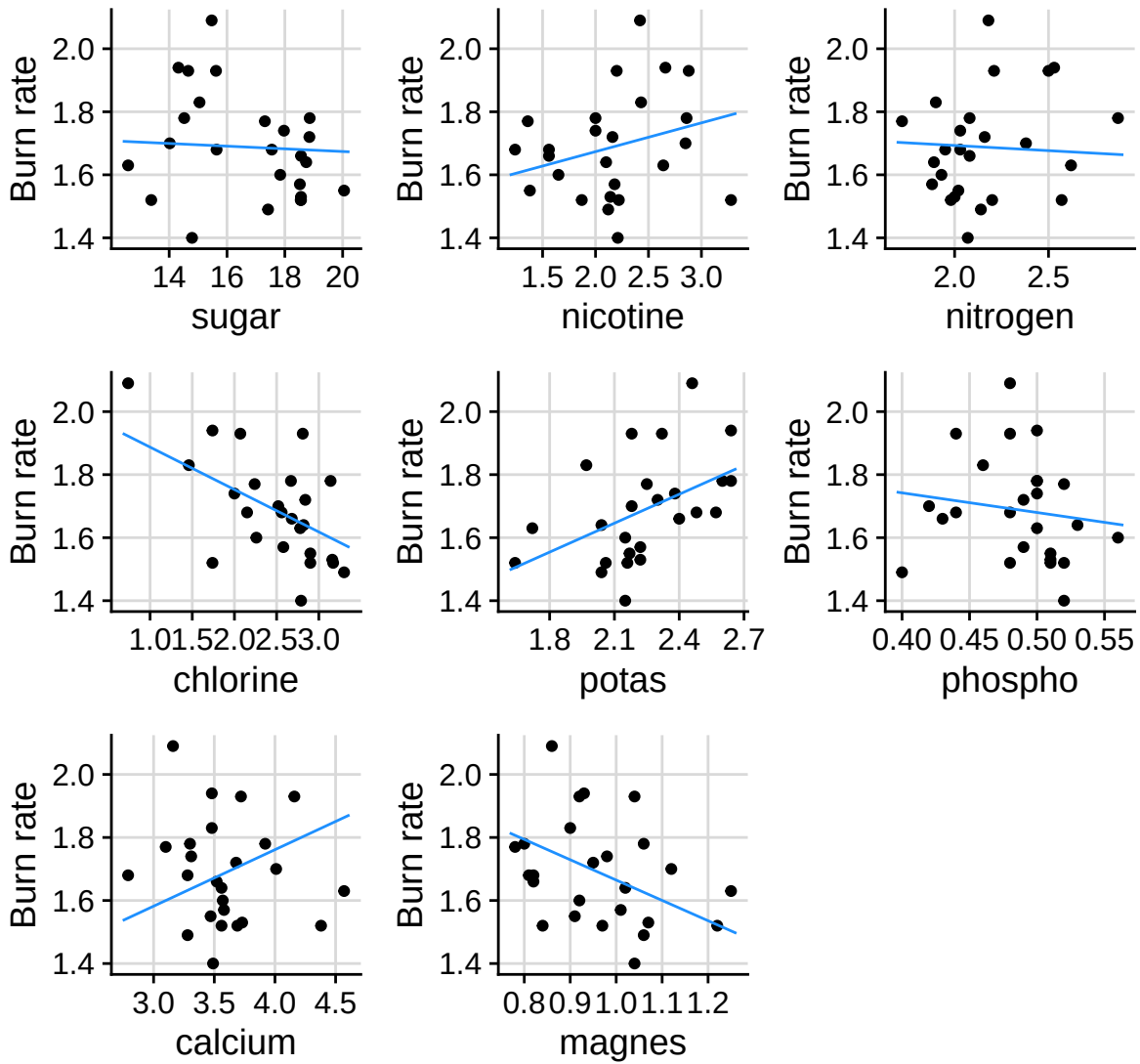
```
# first calculate the mean of every variable
mean_mat <- data_tidy_tobacco |>
  dplyr::summarise(dplyr::across(everything(), mean)) |>
  dplyr::select(-burn) |>
  as.matrix()
p_list <- list()
for (i in seq_len(ncol(mean_mat))) {
  nm_curr <- dimnames(mean_mat)[[2]][i]
  range_vec <- range(data_tidy_tobacco[[nm_curr]])
  range_len <- diff(range_vec)
  x_vec <- range_vec + c(-0.025, 0.025) * range_len
  beta_ind <- which(rownames(beta_vec) == nm_curr)
  beta_mat_rest <- beta_vec[-beta_ind, ] |> as.matrix()
  mean_mat_rest <- mean_mat[, -i, drop = FALSE]
```

```

mean_mat_rest_int <- cbind(1, mean_mat_rest)
y_vec <- rep(mean_mat_rest_int %*% beta_mat_rest, 2) + beta_vec[beta_ind] * x_vec
plot_tbl <- data_tidy_tobacco[, c(nm_curr, "burn")]
colnames(plot_tbl) <- c("x", "y")
p <- ggplot(
  tibble::tibble(x = x_vec, y = y_vec)
) +
  cowplot::theme_cowplot() +
  cowplot::background_grid(major = "xy") +
  geom_point(mapping = aes(x = x, y = y), data = plot_tbl) +
  geom_line(aes(x = x, y = y), col = "dodgerblue") +
  labs(x = nm_curr, y = "Burn rate")
p_list <- p_list |> append(list(p))
}
p_grid <- cowplot::plot_grid(
  plotlist = p_list,
  ncol = 3,
  nrow = 3
)
path_p_grid <- "_tmp/p-grid.pdf"
cowplot::ggsave2(
  filename = path_p_grid,
  plot = p_grid,
  units = "cm",
  height = 15,
  width = 16
)

pander::pandoc.image(path_p_grid)

```



Part b

Do so using built-in R functions (i.e. `lm`).

Sub-part i

Display the summary output (use the `summary` object on the fitted model object above).

```
fit <- lm(burn ~ ., data = data_tidy_tobacco)
summary(fit)
```

Call:

```
lm(formula = burn ~ ., data = data_tidy_tobacco)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-0.149403	-0.048228	0.001326	0.047505	0.137921

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.563675	0.712327	2.195	0.0432 *
sugar	-0.004242	0.019449	-0.218	0.8301
nicotine	0.091315	0.095146	0.960	0.3515
nitrogen	-0.032790	0.191465	-0.171	0.8662
chlorine	-0.134570	0.056919	-2.364	0.0310 *
potas	0.305924	0.168749	1.813	0.0887 .
phospho	-0.623569	0.647028	-0.964	0.3495
calcium	0.179185	0.119419	1.500	0.1530
magnes	-0.645189	0.404667	-1.594	0.1304

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1077 on 16 degrees of freedom

Multiple R-squared: 0.7226, Adjusted R-squared: 0.5838

F-statistic: 5.209 on 8 and 16 DF, p-value: 0.002488